

Identification of anthocyanins in the liver, eye, and brain of blueberry-fed pigs.

Dietary intervention with anthocyanins may confer benefits in brain function, including vision. Research to date indicates that animals have only a limited capacity to absorb anthocyanins, compared to other types of flavonoids. Pigs, which are a suitable model for human digestive absorption, were used to examine the deposition of anthocyanins in tissues including the liver, eye, and brain tissue. Pigs were fed diets supplemented with 0, 1, 2, or 4% w/w blueberries (*Vaccinium corymbosum* L. 'Jersey') for 4 weeks. Prior to euthanasia, pigs were fasted for 18-21 h. Although no anthocyanins were detected in the plasma or urine of the fasted animals, intact anthocyanins were detected in all tissues where they were sought. LC-MS/MS results are presented for the relative concentration of 11 intact anthocyanins in the liver, eye, cortex, and cerebellum. The results suggest that anthocyanins can accumulate in tissues, including tissues beyond the blood-brain barrier.